

Name _____

The Sky is the Limit — Decoding Weather Conditions

Student Answer Packet 2: To be completed individually. You may use information from **Student Directions** and **Student Answer Packet 1** to help you answer some of these questions.

QUESTIONS:

1. From your investigation of the two different pressure blocks, explain why one blue block (parcel of cold air) is more dense than one red block (parcel of warm air).

2. Describe the general relationship between altitude and air pressure by using **one** piece of numerical evidence from the *Vertical Cross Section of Air Pressure* (at 7 a.m. December 9, 2010) in a *Column of Warm and Cold Air* in the **Weather Investigation Packet** to support your answer.

- 3a. Describe how the movement of the cold front shown on the *Wednesday, December 14, 2011* surface weather map and the *Thursday, December 15, 2011* surface weather map resulted in changes to **two** weather conditions in Abilene, Texas. Use numerical evidence from the maps to support your answer.

- 3b. Identify **one** piece of evidence from the station models found on the *Thursday, December 15, 2011* surface weather map that describes how the movement and interactions of the air masses located over the eastern half of the United States associated with the low-pressure system changed weather conditions compared to the weather conditions on the previous day (Wednesday, December 14, 2011).

4. Using the *Planetary Wind Belts in the Troposphere Model* found in the ***Weather Investigation Packet***, identify the name of the wind belt that influences the weather conditions across the continental United States, and describe the evidence for how this global wind pattern influenced the movement of weather systems from December 14 to December 15, 2011.

Name of global wind belt: _____

Description: _____

5. Use the satellite images in the ***Weather Investigation Packet*** and your knowledge of weather to describe the general circulation patterns around the center of Hurricane Andrew and Tropical Cyclone (Hurricane) Orson. Include **two** descriptions of the general circulation pattern for **each** hurricane.

Hurricane Andrew: _____

Tropical Cyclone (Hurricane) Orson: _____

6. Use the *Global Surface Ocean Currents Map* in the ***Weather Investigation Packet*** to explain why hurricanes do **not** form in the Pacific Ocean along the west coast of the United States.

7. Identify a date and specific location from **one** of the surface weather maps that can be used to infer where a snow event may occur. Also identify two pieces of evidence from the map to support your choice.

Selected date: _____

Location: _____

Evidence:

1. _____

2. _____

- 8a. How do atmospheric density differences in columns of warm and cold air influence the interaction of different air masses at frontal boundaries?

- 8b. Explain why there is a lack of precipitation behind the cold front that extends from Canada to Texas on the *Thursday, December 15, 2011* surface weather map, and explain why precipitation is occurring in front of the cold front. Include the concept of density in your answer.

9. Describe how the position of the northern hemisphere jet stream influenced the seasonal shift in the temperature gradient values for the two boxed locations found on the surface weather maps on *Wednesday, December 14, 2011* and *Saturday, June 16, 2012*.

- 10a. Use the surface weather maps and the data from question 15 on ***Student Answer Packet 1*** to describe how the characteristics of the air masses over Albany, NY, changed during the three-day period from *Wednesday, December 14, 2011* through *Friday, December 16, 2011*.

- 10b. Explain how air mass geographic locations influence the characteristics associated with an air mass.
